

HOLOCRON

Project Overview

Holocron is a private, secure, and customizable LLM infrastructure aimed at enterprises. The system allows organizations to capture and leverage their proprietary data, offering full control over the AI pipeline and enabling enhanced intelligence capabilities. Inspired by the knowledge-preserving Holocrons from *Star Wars*, this system integrates AI with organizational data to help businesses generate insights, optimize workflows, and improve user experiences.

Goals & Capabilities

- Build a proprietary LLM tailored to client's data and use cases
- Enable voice/text queries with advanced contextual understanding
- Ensure HIPAA-grade compliance, security, and auditability
- Seamless integration with enterprise tools (Slack, CRMs, etc.)
- Advanced ticketing, knowledge base, and automation support

Phase 1 – Data Ingestion & Preprocessing

This phase focuses on securely ingesting and preparing multi-format organizational data for embedding and training, laying the groundwork for intelligent retrieval and contextual analysis.

- **Input Support:** Accepts PDFs, Word docs, Excel files, JSON, CSV, and website text
- **Preprocessing Pipeline:**
 - Extracts text using parsing engines (PDFMiner, Python-docx, Pandas, etc.)
 - Normalizes content for consistent formatting
 - Annotates metadata such as source, tags, timestamps
- **Sensitive Data Masking:**
 - Automatically detects and redacts PII (emails, SSNs, phone numbers)
 - Enforces compliance with HIPAA, GDPR
- **Multi-source Integration:**

- Supports ingestion from Remedy tickets, structured logs, and public web content
- **Diversity & Quality Assurance:**
 - Performs noise removal, deduplication, and bias checks
 - Ensures representation of all relevant query types and content categories

Phase 2 – Embedding Generation & FAISS Indexing

This phase converts the processed text into semantically searchable embeddings and indexes them for high-speed retrieval.

- **Embedding Model:** Uses all-MiniLM-L6-v2 or equivalent to convert text into high-dimensional vector format
- **FAISS Indexing:**
 - Stores embeddings in a similarity-search-optimized FAISS index
 - Associates each entry with document metadata for contextual relevance
- **Metadata Storage:**
 - Titles, source identifiers, document type, and tags stored alongside embeddings
- **Real-Time Scalability:**
 - Index designed to handle millions of records with constant-time retrieval
 - Supports updates and versioning of indexed data

Phase 3 – Query Processing & Reranking

This stage handles incoming user queries (text or voice), performs semantic search, and prepares results for LLM reasoning.

- **Query Embedding:**
 - User queries converted into embeddings using the same model used during indexing
- **Similarity Search:**
 - Retrieves top matching documents from FAISS index

- Ensures fast, relevant retrieval based on meaning rather than keywords
- **Reranking:**
 - Applies ms-marco-MiniLM-L-6-v2 cross-encoder to improve relevance
 - Selects Top-N results (e.g., top 4) for final answer synthesis

Phase 4 – Answer Generation & Output Delivery

In this final phase, the retrieved context is merged with the user's query to generate structured, conversational outputs.

- **LLM Integration:**
 - Uses OpenAI GPT-4 or fine-tuned internal model to generate responses
 - Can summarize, explain, or answer based on combined query + context
- **Voice & Text Interface:**
 - Voice-to-text and text-to-voice capabilities for multimodal access
 - Adaptive voice recognition for natural conversation patterns
- **API Response Output:**
 - JSON output structure includes original sources, confidence scores, and result context
 - Supports integration with ticketing, CRM, and ERP systems
- **Audit & Feedback:**
 - Logs user queries, output history, and reranking results for improvement
 - Admin review tools for refining edge cases and boosting quality

Technology Stack

The Holocron platform is built using a secure, modular, and cloud-native architecture, tailored specifically for enterprise knowledge management and intelligent automation. The technologies have been carefully selected to ensure scalability, real-time semantic search, multi-modal interaction, and enterprise-grade security.

Voice Interface

Holocron enables users to interact with the system through both voice and text, creating a flexible and accessible experience across use cases.

- Whisper is used for real-time speech-to-text transcription to convert verbal prompts into structured text.
- Amazon Polly or Google TTS is used to synthesize natural-sounding voice responses in multiple languages, including English, Hindi, and Spanish.
- Voice interaction is context-aware and supports adaptive recognition across accents and languages.

Voice & Chat Integration

- Holocron supports real-time voice interaction on mobile, web, and assistant devices via WebRTC or SIP-based voice API.
- Chat-based interface is supported with persistent context memory and user history.
- For enterprise deployments, Holocron integrates with ticketing systems, CRM chatbots, and internal Slack/MS Teams bots.

Backend System

- FastAPI powers the backend, offering an async, high-performance framework for managing user queries, file ingestion, and API workflows.
- PostgreSQL (AWS RDS) stores all document metadata, user sessions, feedback logs, and access records.
- Celery/RQ enables background job handling for tasks like document ingestion, embedding generation, and similarity scoring.
- FAISS is used as the vector search engine to support high-speed, semantically relevant document retrieval.

Frontend Interface

- Holocron's admin portal is built with React.js, allowing knowledge managers to upload data, monitor query logs, and manage user roles.
- TailwindCSS is used for rapid, responsive styling.
- Vite ensures fast load times, optimized bundling, and smooth developer experience.

DevOps & Cloud Infrastructure

- All components are containerized via Docker and deployed on AWS ECS / Fargate.
- S3 is used for document storage, logs, and versioned embeddings.
- API Gateway governs secure external access, including webhooks and client integrations.
- CloudWatch monitors application health and logs ingestion, query traffic, and voice activity.
- IAM + VPC is used for fine-grained permission management and private service networking.

Security Architecture

- All data in transit and at rest is encrypted using TLS 1.3 and AES-256.
- OAuth 2.0 is used for secure, token-based authentication and authorization.
- Role-Based Access Control (RBAC) and session audit trails are enforced to ensure data access integrity and traceability.
- Personally identifiable information (PII) is masked and stored only with user consent or role-based access.

Compliance & Risk Monitoring

- The infrastructure follows SOC2, HIPAA, and GDPR compliance standards.
- Security scanning is automated with CloudSploit and ScoutSuite, enforcing policy and vulnerability checks.
- All data ingestion, processing, and user interactions are logged for full audit traceability.

Team Structure

To successfully deliver the Holocron LLM Platform, Brightcone.ai proposes a lean, highly-skilled technical team specializing in AI systems, vector search, and secure cloud deployments. The team is designed for rapid iteration, efficient integration, and enterprise-grade delivery.

- **2 AI/Backend Engineers**
 - Expertise in FastAPI, vector databases (FAISS), and LLM integration
 - Responsible for ingestion workflows, embedding pipelines, query processing, and API development
- **1 Machine Learning Engineer**
 - Focused on fine-tuning LLMs, optimizing embedding models, and improving semantic search performance
 - Handles evaluation, reranking, and query-answer quality metrics
- **1 DevOps Engineer**
 - Manages containerization, cloud infrastructure (AWS), CI/CD pipelines, and deployment automation
 - Ensures uptime, scalability, and secure access provisioning
- **1 Product Lead / Solution Architect**
 - Oversees project execution, technical decisions, and alignment with client goals
 - Ensures domain customization and compliance alignment

Security Protocols

Given the enterprise and potentially sensitive nature of the knowledge processed by Holocron, robust security and compliance safeguards are enforced at every layer of the architecture.

- **Encryption:**
 - AES-256 encryption at rest
 - TLS 1.3 encryption for all data in transit
- **Authentication & Access Control:**
 - OAuth 2.0-based token authentication
 - Role-Based Access Control (RBAC) for limiting user permissions
 - Multi-Factor Authentication (MFA) supported at the admin level
- **Data Privacy & Retention:**

- Masking of personally identifiable information (PII) in ingestion pipelines
- Temporary staging for intermediate data with automatic expiration
- Explicit deletion endpoints and data lifecycle policies
- **Audit & Compliance:**
 - Full audit logging for every system action (including document ingestion, query execution, API access)
 - Logs include timestamps, user IDs, IP addresses, and action type
 - Infrastructure aligned with SOC 2, HIPAA, and GDPR standards
- **Monitoring & Threat Detection:**
 - Continuous vulnerability scanning using tools such as ScoutSuite and CloudSploit
 - Alerts and anomaly detection via AWS CloudWatch and security monitoring dashboards

Architecture

The Holocron platform is designed with a modular, scalable architecture that supports semantic search, voice interaction, and LLM-powered answer generation. It integrates seamlessly with enterprise systems through APIs and supports asynchronous background processing for document ingestion, embedding, and retrieval.

High-Level Step-by-Step Flow

1. Document Upload & Metadata Registration

- Users upload files (PDFs, DOCX, CSV, etc.) through the admin portal or API.
- The system stores files in AWS S3 and logs metadata (filename, user, source, type) in PostgreSQL (RDS).

2. Trigger Embedding & Indexing (API #1)

- External systems or the portal call:
POST /ingest/<document_id>
- This triggers a background job via Celery/RQ to:
 - Fetch the document from S3

- Preprocess and normalize the text
- Mask sensitive data and annotate with metadata

3. Background Job: Embedding & Indexing

- The cleaned document is passed through the embedding model (e.g., all-MiniLM-L6-v2) to create a semantic vector.
- Embeddings are stored in FAISS, and document metadata is stored in PostgreSQL.
- The system maintains:
 - Version history for document updates
 - State machine status for tracking processing lifecycle:
 - pending
 - preprocessed
 - embedded
 - indexed
 - error
 - ready

4. User Query & Retrieval (API #2)

- User submits a query via chat, voice, or portal:
 - POST /query
- The query is embedded and compared against FAISS vectors.
- Top matches are reranked using ms-marco-MiniLM-L-6-v2.

5. LLM Response Generation

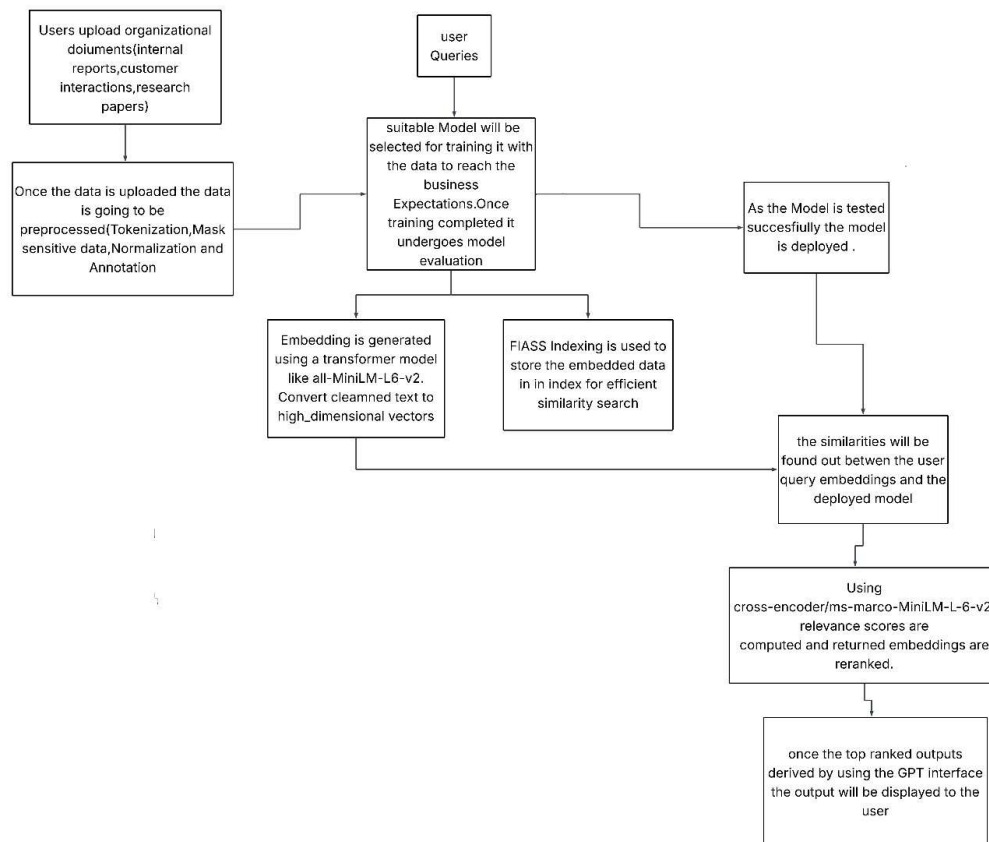
- Top-N results and the user query are passed to OpenAI GPT-4 or a fine-tuned internal model.
- The model generates:
 - Structured responses
 - Summaries or direct answers
 - Citations (links to source documents)

6. API Output Delivery

- Final response is returned as structured JSON, including:
 - Answer
 - Source metadata
 - Confidence scores
 - Flags (e.g., low-confidence, outdated)

7. Optional Feedback Loop

- Users or admins can:
 - Provide feedback to improve future results
 - Trigger reprocessing or correction workflows
 - View query logs, past results, and source tracebacks



Deliverables and Timeline

Sprint	Timeline	Key Deliverables	Estimated Cost
Sprint 1	Weeks 1–2	Initial setup: environment, model architecture, and data pipeline	
		API scaffolding to handle requests and initiate model training tasks	
		Basic integration of the core LLM model (e.g., fine tuning the initial version)	
		Data collection and preprocessing pipeline setup (cleaning and formatting training data)	
Sprint 2	Weeks 3–4	Setup background job processing for data and model training	
		Prototype core LLM model integration (train on a subset of data)	
		API integration for document processing and result retrieval	
		Logging and error handling for model training and document processing	
Sprint 3	Weeks 5–6	Implement multi language support for model output (e.g., French, Spanish)	
		Integrate confidence scoring and result verification for LLM outputs	
		Expand model capabilities to handle additional data types and document structures	
		Unit tests for model components and output verification	
Sprint 4	Weeks 7–8	Implement output formatting for end user use (JSON, CSV, etc.)	
		Link model results to original document metadata records	
		Secure API endpoints for triggering and fetching model results	
		Implement version control and auditing features for model output and processing logs	
Sprint 5	Weeks 9–10	Test and validate system performance with a full scale dataset (load testing)	
		Optimize LLM model for accuracy and performance (fine tuning, hyperparameter tuning)	
		Integrate additional security features (e.g., encryption, access control)	
Sprint 6	Weeks 11–12	Finalize model and system integration for deployment	

		Perform comprehensive system testing (including edge cases and failure scenarios)	
		Final documentation and reporting for project handover	
		Post deployment support plan and monitoring setup	

Monthly Cloud & AI Cost Estimate

Brightcone.AI – Unique Value Proposition